

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY**Eighth Semester of B. Tech (ME) Examination****April - May 2018****ME420 Optimization Techniques (Elective – II)****Date: 09.05.2018, Wednesday****Time: 01.30 p.m. to 04.30 p.m.****Maximum Marks: 70****Instructions:**

1. The question paper comprises two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I**Q-1(a) Answer True or False. (Each carries one mark)****[07]**

- (1) A design point that lies on one or more than one constraint surface is called a 'bond point' and the associated constraint is called 'active constraint'.
- (2) One of the engineering applications of optimization is Inventory control.
- (3) The sequence F_n of Fibonacci numbers is defined by $F_n = F_{n-1} + F_{n-2}$.
- (4) Newton-Raphson method is having comparatively slow computation than Quasi-Newton method.
- (5) In Quasi-Newton method we need to calculate second derivative of the function.
- (6) Steepest – Descent method is a Linear Programming method.
- (7) In Steepest – Descent method, for minimization problem, gradient direction is the direction of steepest ascent.

Q-1(b) Define Inflection / Saddle point and determine minimum / maximum / inflection point of $F(X) = X^3$.**[04]****Q-1(c) Classify optimization problem based on –****[04]**

- (1) The existence of constraints
 - (2) The nature of the equation involved
- And give names of the methods included in each.

Q-2 Find the extreme points of the given functions by using Lagrange Multiplier. (Any two)**[12]**

- (1) $F(X, Y, Z) = X + Y + 2Z$
Subjected to $X^2 + Y^2 + Z^2 = 3$
- (2) $F(X, Y) = XY$
Subjected to $3X^2 + Y^2 = 6$
- (3) $F(X, Y, Z) = X^2 + Y^2 + Z^2$
Subjected to $X + Y + Z = 1, X + 2Y + 3Z = 6$

Q-3 Minimize following functions by using Kuhn-Tucker Conditions. (Any one)**[08]**

- (1) $F = X_1^2 + 2X_2^2 + 3X_3^2$
Subjected to-

$$X_1 - X_2 - 2X_3 \leq 12$$

$$X_1 + 2X_2 - 3X_3 \leq 8$$
- (2) $F = X_1^2 + X_2^2 + 60X_1$
Subjected to-

$$X_1 - 80 \geq 0$$

$$X_1 + X_2 - 120 \geq 0$$

SECTION – II

Q-4 Answer True or False. (Each carries one mark)

[07]

- (1) Exhaustive search method is a multi-variable non-linear programming method.
- (2) Dichotomous method is a single variable linear programming method.
- (3) Fibonacci and Golden section methods are gradient based non-linear programming method.
- (4) Random-Walk method is also called Cauchy's method.
- (5) In Univariate method, we change only one variable at a time and seek to produce a sequence of improved approximations to the minimum point.
- (6) In Golden section method, golden section (γ) = 1.618.
- (7) fminunc command is used to find minimum of unconstrained multivariable function in MATLAB.

Q-5 (a) Minimize following function by Random-Walk method OR by Steepest-Decent method up to two iterations. **[08]**

$$F(X_1, X_2) = 4X_1^2 + 3X_2^2 - 6X_1X_2 - 4X_1$$

Subjected to -

$$-10.0 \leq X_1, X_2 \leq 10.0$$

$$\text{Take } \lambda = 1.2, \epsilon = 0.075, N = 2.$$

Q-5 (b) Minimize following function by using Powell's method. **[10]**

$$F(X_1, X_2) = X_1 - X_2 + 2X_1^2 + 2X_1X_2 + X_2^2$$

From the starting point $X_1 = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$. Take $\epsilon = 0.01$.

Q-6 Answer following questions. (Any two)

[10]

- (1) Find the minimum of $F = X(X^3 - 2)$ in the interval (0, 1.5) to within 16% of the exact value by Exhaustive Search method.
- (2) Explain the general procedure to find the minimum of the function by Secant method.
- (3) Explain the general procedure to find the minimum of the function by Quadratic Interpolation method.
